



Cambridge (CIE) IGCSE Computer Science



Your notes

The Internet & the World Wide Web

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Your notes

The Internet & the World Wide Web

The Internet & the World Wide Web

What is the Internet?

- The Internet is a **global network of networks**
- The Internet is the most well-known Wide Area Network (**WAN**)
- The Internet is the **infrastructure** used to **provide connectivity** to the World Wide Web

What is the World Wide Web?

- The world wide web, or simply **the web**, is a **collection of websites** and **web pages** that are **accessed using the internet**
- It was created in 1989 by **Tim Berners-Lee**, who envisioned it as a way to **share** and **access** information **on a global scale**
- The web consists of **interconnected documents** and **multimedia files** that are stored on **web servers** around the world
- **Web pages** are accessed using a web browser, which communicates with web servers to retrieve and display the content

Network Protocols

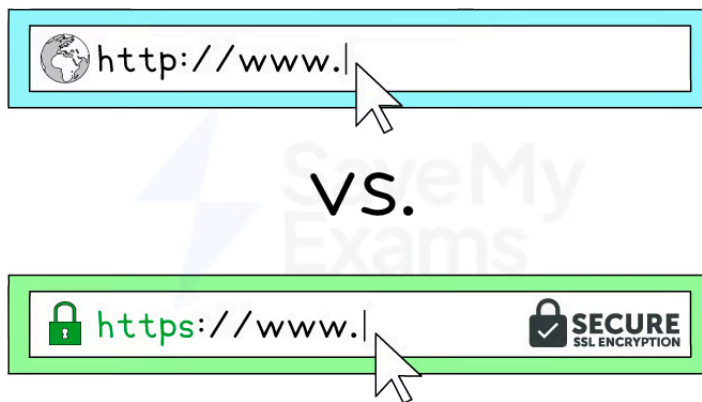


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HTTP & HTTPS

What is HTTP & HTTPS?

- Hypertext Transfer Protocol (**HTTP**) allows communication between clients and servers for **website viewing**
- HTTP & HTTPS are **protocols**, a **set of rules** governing communication between devices on a network
- HTTP allows clients to **receive data** from the sever (fetching a webpage) and **send data** to the server (submitting a form, uploading a file)
- **HTTPS** works in the same way as HTTP but with an **added layer of security**
- All data sent and received using HTTPS is **encrypted**
- HTTPS is used to **protect sensitive information** such as passwords, financial information and personal data



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Web Browser

Web Browser

What is a web browser?

- A web browser is a **piece of software** used to **access** and **display** information on the internet
- A web browser **displays web pages** by **rendering hypertext markup language (HTML)**
- Web browsers **interpret the code in HTML** documents and **translate** it into a visual display for the user

Functions of a web browser

Function	Description
Render HTML	Display the web page
Provide navigation tools	E.g. back/forward buttons and home button, to help users move between pages
Storing bookmarks & favourites	Allow users to save links to frequently visited websites and access them easily
Storing cookies	Cookies
Record user history	Allow users to quickly revisit recently viewed web pages
Provide address bar	A place for user to type in the URL (link to URL page) of a web page to visit
Multiple tabs	Allow multiple web pages to be open at once so users can quickly switch between them



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Web Pages

Loading a Web Page

How is a web page loaded?

- Web pages **are held on web servers** (1), known as '**hosting**'
- To access a web page on a web server, **a web browser is used**
- In the browser, a user enters **a web page URL** (2)
- The browser sends the **domain name** to a **DNS** (3)
- The browser connects to the web server and **requests to access the page**
- **HTML** (4) is transferred and **rendered by the browser**, displaying the web page

Web Servers (1)

What is a web server?

- A web server is **a remote computer** that **stores the files needed to display a web page** on the Internet
- Web servers are generally **available 24/7** and **security is managed** by the owner of the hardware
- Web servers provide access to **multiple users at the same time**

Uniform Resource Locator (URL) (2)

What is a URL?

- A **Uniform Resource Locator** (URL) is a **unique identifier** for a web page, known as the **website address**
- It is **text based** to make it easier to remember
- A user **enters a URL** into a web browser to view a web page
- An example of a URL is:

<https://www.savemyexams.com/igcse/computer-science/cie/23/revision-notes/>

- A URL can typically be split into three parts:
 - **Protocol**



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- Domain name
- Web page/file name
- Using the example about the URL would be split as follows:

Protocol	https	Communication method to transfer data between client and server
Domain name	www.savemyexams.com	Name of the server where the resource is located
Web page/file name	/igcse/computer-science/cie/23/revision-notes/	Location of the file or resources on the server

Domain Name System (DNS) (3)

What is a DNS?

- The Domain Name System (DNS) can be thought of as the Internet's equivalent to a **phone book**
- It is essentially a directory of domain names and is used to translate human-readable domain names to the numeric **IP addresses** that computers use
- When you type a URL into your browser, **the DNS translates the domain name into its associated IP address** so your computer can connect to the server hosting the website
- Without DNS, we would have to remember the IP address of every site we want to visit

HTML (4)

What is HTML?

- Hypertext Markup Language (**HTML**), is the foundational language **used to structure and present content on the web**
- HTML consists of a series of elements, often referred to as "**tags**"
- Most tags are **opened and closed e.g. <html> and </html>**, whereas **some tags are only opened e.g. and <link>**

Structure

- HTML is used to **define the basic structure of a webpage** by organising content into **sections** such as headers, paragraphs, and footers



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- The `<html>` tag is the **root element** of an HTML page and includes all other HTML elements used to create a page structure

```
<><!DOCTYPE html>
<html>
  <head>
    <title>My Web Page</title>
  </head>
  <body>
    <header>
      <h1>Welcome to My Website</h1>
    </header>
    <main>
      <section>
        <h2>About Me</h2>
        <p>This is a paragraph about me.</p>
      </section>
      <section>
        <h2>My Projects</h2>
        <p>This is a paragraph about my projects.</p>
      </section>
    </main>
    <footer>
      <p>Contact: myemail@example.com</p>
    </footer>
  </body>
</html>
```

- In this example, HTML is used to create a structure with a **header**, two sections in the **main body**, and a **footer**
- Other examples of HTML being used for structure include:
 - **Creating lists to structure information**
 - **Positioning of text on the screen**
 - **Embedding media and interactive elements**

Present

- HTML is also used to **present and display information in a visually** meaningful way
- The content layer of a web page is made up of HTML elements such as **headings** (`<h1>`, `<h2>`, etc.), **paragraphs** (`<p>`), **links** (`<a>`), **images** (`<img>`), and more
- This layer is mainly handled by **CSS** (Cascading Style Sheets)

```
<><!DOCTYPE html>
<html>
  <body>
    <h1>Welcome to My Website</h1>
```



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<p>This is a paragraph introducing the content of the website.</p>

<h2>Subheading 1</h2>

<p>Here is some detailed information under the first subheading.</p>

<h2>Subheading 2</h2>

<p>Another section with more information.</p>

<p>Bold text and italic text can emphasise important points.</p>

</body>

</html>

- In this example, **headings** (<h1>, <h2>) and **text formatting** tags (,) are used to present the content clearly and with emphasis
- Other examples of HTML being used to present information include:
 - **Presenting data in a table**
 - **Displaying images with captions**



Worked Example

A company sells products over the Internet.

Explain how the information stored on the company's website is requested by the customer, sent to the customer's computer and displayed on the screen.

[7]

Answer

Seven from:

Requested

- a web browser is used
- user enters the URL / web address (into the address bar) // clicks a link containing the web address // clicks an element of the webpage
- the URL / web address specifies the protocol
- protocols used are Hyper Text Transfer Protocol (HTTP) / Hyper Text Transfer Protocol Secure (HTTPS)

Sent

- the URL / web address contains the domain name
- the domain name is used to look up the IP address of the company
- the domain name server (DNS) stores an index of domain names and IP addresses

- web browser sends a request to the web server / IP address

Received

- Data for the website is stored on the company's web server
- webserver sends the data for the website back to the web browser
- web server uses the customer's IP address to return the data
- the data is transferred into Hyper Text Mark-up Language (HTML)
- HTML is interpreted/rendered by the web browser (to display the website)



Your notes



Your notes

Cookies

Cookies

What is a cookie?

- A cookie is a **tiny data file** stored on a computer by browser software that holds information relating to your **browsing activity**
- Typically a cookie will contain:
 - **Browsing history** - what websites you have visited
 - **Login information** - usernames & passwords
 - **Preferences** - language/font size/themes
- The two types of cookie are:
 - **Session** - Created and replaced every time a user visits a website
 - **Persistent** - Created and saved the first time a user visits a website, retained until they expire

What is the Privacy and Electronic Communications Regulations (2003)?

- The Privacy and Electronic Communications Regulations (2003) is a law that **governs the use of cookies**
- Any website that wants to store a cookie must:
 - **Tell users the cookies are there**
 - **Explain what the cookies are doing**
 - **Obtain users consent to store the cookie**